

SRM VEC

SRM Nagar, Kattankulathur – 603 203

DEPARTMENT OF AIDS

QUESTION BANK

SUBJECT: AD3445 – AI1

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UNIT I – INTRODUCTION TO AI AND PROBLEM SOLVING

History of AI, Intelligent Agents, Agents and Environments, Nature of Environments, Structure of Agents, Problem Solving Agents, Searching for Solutions, Uninformed Search Strategies: Breadth-first search, Depth-first search, Depth-limited search, Iterative deepening depth-first search, Bidirectional search

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define Artificial Intelligence according to the 'Thinking Humanly' approach.	BTL1	Remembering
2	What is the Turing Test and what is its primary objective?	BTL1	Remembering
3	List the four components of a PEAS description for an intelligent agent.	BTL1	Remembering
4	Distinguish between a fully observable and a partially observable environment.	BTL2	Understanding
5	Define a 'Rational Agent' in the context of AI.	BTL1	Remembering
6	What is the difference between an 'Agent Program' and an 'Agent Architecture'?	BTL2	Understanding
7	Identify the characteristics of a 'Stochastic' environment.	BTL2	Understanding
8	Mention the four criteria used to evaluate the performance of search strategies.	BTL1	Remembering
9	Define 'State Space' and 'Path Cost' in a problem-solving context.	BTL1	Remembering
10	Why is Breadth-First Search (BFS) considered optimal for problems with unit step costs?	BTL2	Understanding
11	State the main disadvantage of Depth-First Search (DFS) regarding completeness.	BTL2	Understanding
12	What is the core idea behind Iterative Deepening Depth-First Search (IDDFS)?	BTL2	Understanding
13	Under what condition is Bidirectional Search not applicable?	BTL2	Understanding
14	Define 'Depth-Limited Search' and explain how it addresses the issues of DFS.	BTL1	Remembering
15	What is a 'Simple Reflex Agent' and what are its limitations?	BTL1	Remembering
16	Contrast 'Static' and 'Dynamic' environments with an example for each.	BTL2	Understanding
17	What is the significance of the 'Goal Test' in a search algorithm?	BTL2	Understanding
18	Define 'Branching Factor' and 'Depth' of a search tree.	BTL1	Remembering
19	What is an 'Actuator' in the context of an intelligent agent?	BTL1	Remembering
20	Explain the role of a 'Utility Function' in a utility-based agent.	BTL2	Understanding

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Trace the evolution of Artificial Intelligence from its inception at the Dartmouth conference to the current era, highlighting key milestones and the impact of 'AI Winters'.	(16)	BTL4	Analyzing
2	Design a comprehensive PEAS description and analyze the environment characteristics for an 'Automated Taxi Driver' and a 'Medical Diagnosis System'.	(16)	BTL6	Creating
3	Compare and contrast the five basic types of agent programs (Simple Reflex, Model-based, Goal-based, Utility-based, and Learning agents) with suitable diagrams and examples.	(16)	BTL4	Analyzing
4	Formulate the 8-puzzle problem as a formal state-space search problem, clearly defining the states, initial state, actions, transition model, and goal test.	(16)	BTL3	Applying
5	Explain the Breadth-First Search (BFS) algorithm in detail. Provide a proof for its completeness and analyze its time and space complexity using Big-O notation.	(16)	BTL4	Analyzing

6	Critically evaluate the performance of Depth-First Search (DFS) versus Depth-Limited Search (DLS) when applied to an infinite state space. Which is more reliable and why?	(16)	BTL5	Evaluating
7	Illustrate the working of Iterative Deepening Depth-First Search (IDDFS) with a step-by-step example on a search tree. Explain why it is preferred over BFS and DFS in large search spaces.	(16)	BTL3	Applying
8	Analyze the requirements and computational challenges of implementing a Bidirectional Search. Under what specific conditions does it significantly outperform standard BFS?	(16)	BTL4	Analyzing
9	Develop a model-based reflex agent for a 'Vacuum-Cleaner World'. Explain how the internal state helps the agent overcome the limitations of a partially observable environment.	(16)	BTL6	Creating
10	Compare the following environment types with real-world examples: (i) Discrete vs Continuous, (ii) Single-agent vs Multi-agent, (iii) Known vs Unknown.	(16)	BTL4	Analyzing
11	A robot needs to navigate a 5x5 grid from (0,0) to (4,4). Apply the logic of Uniform Cost Search (as a variant of uninformed search) to find the path, assuming diagonal moves cost 1.4 and orthogonal moves cost 1.0.	(16)	BTL3	Applying
12	Discuss the concept of 'Rationality' and 'Bounded Rationality'. How does the nature of the environment (e.g., time constraints) affect the rationality of an agent's decision-making?	(16)	BTL5	Evaluating
13	Construct a search tree for a navigation problem between two cities. Demonstrate the execution of Depth-First Search, showing the state of the frontier and explored set at each step.	(16)	BTL3	Applying
14	Explain the architecture of a Learning Agent. How do the 'Performance Element', 'Learning Element', 'Critic', and 'Problem Generator' interact to improve agent performance over time?	(16)	BTL4	Analyzing
15	Justify the use of Iterative Deepening DFS over BFS in memory-constrained systems. Provide a mathematical comparison of their space complexities for a tree with branching factor 'b' and depth 'd'.	(16)	BTL5	Evaluating
16	Design a goal-based agent for an automated warehouse robot. Describe its components and explain how it handles a sequence of actions to achieve the goal of delivering a package.	(16)	BTL6	Creating
17	Analyze the impact of the branching factor 'b' and depth 'd' on the time complexity of uninformed search algorithms. How does an exponential complexity limit the practical application of these algorithms?	(16)	BTL4	Analyzing
18	Evaluate the effectiveness of the Turing Test as a measure of machine intelligence. Discuss the 'Chinese Room' argument as a critique of the test's validity.	(16)	BTL5	Evaluating
19	For a 'Chess Playing' agent, define the PEAS, classify the environment type, and propose the most suitable agent structure with a detailed justification for your choice.	(16)	BTL6	Creating
20	Compare BFS, DFS, DLS, and IDDFS based on completeness, optimality, time complexity, and space complexity. Present your analysis in a structured tabular format.	(16)	BTL4	Analyzing

UNIT II – INFORMED SEARCH AND GAME PLAYING

Informed Search Strategies: Greedy best-first search, A search, Memory-bounded heuristic search, Heuristic Functions, Local Search Algorithms and Optimization Problems, Evolutionary Algorithms, Constraint Satisfaction Problems, Game Playing: Optimal Decisions in Games, Alpha-Beta Pruning, Stochastic Games*

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define a heuristic function and state its role in informed search.	BTL1	Remembering
2	Distinguish between admissible and consistent heuristics.	BTL2	Understanding
3	What is the main advantage of Greedy Best-First Search over Uniform Cost Search?	BTL2	Understanding
4	State the condition under which A* search is guaranteed to be optimal.	BTL1	Remembering
5	Define the 'horizon effect' in game playing.	BTL1	Remembering
6	What is the purpose of the 'temperature' parameter in Simulated Annealing?	BTL2	Understanding
7	List the three components that define a Constraint Satisfaction Problem (CSP).	BTL1	Remembering
8	Explain the concept of arc consistency in CSPs.	BTL2	Understanding
9	Define a zero-sum game in the context of AI.	BTL1	Remembering
10	What is the time complexity of Alpha-Beta pruning in the best-case scenario?	BTL1	Remembering
11	Briefly explain the 'mutation' operator in Genetic Algorithms.	BTL2	Understanding
12	What is the primary limitation of Hill Climbing search?	BTL2	Understanding
13	Define 'Stochastic Games' and give one example.	BTL1	Remembering
14	How does Recursive Best-First Search (RBFS) manage memory?	BTL2	Understanding
15	What is a 'Fitness Function' in Evolutionary Algorithms?	BTL1	Remembering
16	Define the term 'Transposition Table' used in game tree search.	BTL1	Remembering
17	Explain the difference between global and local optima in optimization problems.	BTL2	Understanding
18	What is the role of the 'Alpha' and 'Beta' values in Alpha-Beta pruning?	BTL2	Understanding
19	Mention two memory-bounded heuristic search algorithms.	BTL1	Remembering
20	What is the significance of the 'Open List' and 'Closed List' in A* search?	BTL2	Understanding

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Explain the A* search algorithm in detail. Prove that A* is optimal if the heuristic used is admissible.	(16)	BTL4	Analyzing
2	Compare and contrast Greedy Best-First Search and A* Search with respect to completeness, optimality, and complexity.	(16)	BTL4	Analyzing
3	Describe the Simulated Annealing algorithm. Discuss how the cooling schedule affects the probability of finding a global optimum.	(16)	BTL3	Applying
4	Explain the Genetic Algorithm framework. Illustrate the process of selection, crossover, and mutation using the 8-Queens problem.	(16)	BTL3	Applying
5	Analyze the Minimax algorithm for game playing and demonstrate its execution on a game tree of depth 3.	(16)	BTL4	Analyzing
6	Develop the Alpha-Beta pruning algorithm and explain how it improves the efficiency of the Minimax search without changing the result.	(16)	BTL6	Creating

7	Formulate the Map Coloring problem as a Constraint Satisfaction Problem and explain the steps to solve it using Backtracking search with Forward Checking.	(16)	BTL6	Creating
8	Discuss the Recursive Best-First Search (RBFS) and Simplified Memory-Bounded A* (SMA*) algorithms. Evaluate their memory management strategies.	(16)	BTL5	Evaluating
9	Explain the concept of 'Heuristic Functions'. Discuss the methods for deriving admissible heuristics from relaxed problems.	(16)	BTL4	Analyzing
10	Analyze the challenges of playing Stochastic Games. Explain the Expectiminimax algorithm and how it incorporates chance nodes.	(16)	BTL4	Analyzing
11	Evaluate the performance of Hill Climbing search and its variants: Stochastic Hill Climbing, First-choice Hill Climbing, and Random-restart Hill Climbing.	(16)	BTL5	Evaluating
12	Design a CSP-based solution for the Cryptarithmic problem 'SEND + MORE = MONEY'. Explain the constraints and variables involved.	(16)	BTL6	Creating
13	Compare Local Beam Search with standard Hill Climbing. Explain how 'Stochastic Beam Search' helps in maintaining diversity.	(16)	BTL4	Analyzing
14	Discuss the impact of heuristic accuracy on the performance of A* search. Explain the concept of 'Effective Branching Factor'.	(16)	BTL5	Evaluating
15	Explain the AC-3 algorithm for constraint propagation. Demonstrate how it reduces the search space in a CSP.	(16)	BTL3	Applying
16	Evaluate the use of Evaluation Functions in games like Chess. What features are typically considered to estimate the utility of a non-terminal state?	(16)	BTL5	Evaluating
17	Describe the IDA* (Iterative Deepening A*) algorithm. Analyze its space and time complexity compared to standard A*.	(16)	BTL4	Analyzing
18	Explain the Tabu Search optimization technique and how it uses memory to avoid cycles and escape local optima.	(16)	BTL3	Applying
19	Compare the effectiveness of Minimax search in perfectly observable games versus stochastic games. Justify the need for different approaches.	(16)	BTL5	Evaluating
20	Design an evolutionary strategy to solve the Traveling Salesperson Problem (TSP). Define the chromosome representation and fitness function.	(16)	BTL6	Creating

UNIT III – KNOWLEDGE REPRESENTATION AND LOGIC

Knowledge-Based Agents, Logic, Propositional Logic, First-Order Logic: Syntax and Semantics, Knowledge Engineering in First-Order Logic, Inference in First-Order Logic: Propositional vs. First-Order Inference, Unification, Forward Chaining, Backward Chaining, Resolution, Ontological Engineering

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define a Knowledge-Based Agent and its primary components.	BTL1	Remembering
2	Distinguish between the 'TELL' and 'ASK' operations in a knowledge base.	BTL2	Understanding
3	What is meant by 'Entailment' in formal logic?	BTL1	Remembering
4	Explain the difference between a sound inference and a complete inference.	BTL2	Understanding
5	Define the syntax of Propositional Logic.	BTL1	Remembering
6	What is a 'Model' in the context of logical semantics?	BTL2	Understanding
7	List the two types of quantifiers used in First-Order Logic.	BTL1	Remembering
8	Explain the concept of 'Ground Term' in First-Order Logic.	BTL2	Understanding
9	What is the purpose of the Unification process in inference?	BTL1	Remembering
10	Define a Horn Clause and explain its significance.	BTL2	Understanding
11	What is Skolemization in the context of FOL?	BTL1	Remembering
12	Differentiate between Forward Chaining and Backward Chaining at a high level.	BTL2	Understanding
13	State the Resolution principle for Propositional Logic.	BTL1	Remembering
14	What is 'Ontological Engineering'?	BTL1	Remembering
15	Explain the role of an 'Upper Ontology'.	BTL2	Understanding
16	Define 'Substitution' as applied in First-Order Logic inference.	BTL1	Remembering
17	What is the difference between a function and a predicate in FOL?	BTL2	Understanding
18	List the steps involved in the Knowledge Engineering process.	BTL1	Remembering
19	Explain the concept of 'Validity' and 'Satisfiability' in logic.	BTL2	Understanding
20	Define 'Literal' and 'Clause' as used in Resolution.	BTL1	Remembering

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Explain the architecture of a Knowledge-Based Agent. Describe how the agent maintains its internal state and uses logic to make decisions in a dynamic environment.	(16)	BTL3	Applying
2	Analyze the limitations of Propositional Logic in representing complex domains and justify why First-Order Logic is more expressive for real-world applications.	(16)	BTL4	Analyzing
3	Demonstrate the process of converting the following First-Order Logic sentences into Conjunctive Normal Form (CNF), detailing each step: (" x)([P(x) !' Q(x)] "' [¬Q(x) "(R(x)]).	(16)	BTL3	Applying
4	Compare and contrast Forward Chaining and Backward Chaining algorithms. Under what conditions is one preferred over the other in an expert system?	(16)	BTL4	Analyzing
5	Evaluate the Resolution inference rule. Prove its soundness and discuss why it is considered a refutation-complete inference procedure.	(16)	BTL5	Evaluating
6	Design a First-Order Logic Knowledge Base for a 'Library Management System'. Include rules for borrowing limits, overdue fines, and book availability.	(16)	BTL6	Creating

7	Explain the Unification algorithm in detail. Provide a step-by-step trace to unify the literals: $P(a, x, f(g(y)))$ and $P(z, h(z, w), f(w))$.	(16)	BTL3	Applying
8	Analyze the process of Knowledge Engineering. Discuss the challenges faced during the 'Conceptualization' and 'Formalization' phases when building a domain-specific KB.	(16)	BTL4	Analyzing
9	Critically assess the role of Ontological Engineering in the development of the Semantic Web. How do ontologies facilitate interoperability between different systems?	(16)	BTL5	Evaluating
10	Develop a Resolution-based proof to show that 'John is a criminal' given the facts: (1) It is a crime for an American to sell weapons to hostile nations. (2) Nono has some missiles. (3) All of its missiles were sold to it by John. (4) John is American. (5) Nono is a hostile nation.	(16)	BTL6	Creating
11	Explain the semantics of First-Order Logic by describing how interpretations, domains, and assignments are used to determine the truth of a formula.	(16)	BTL3	Applying
12	Analyze the difference between Propositional Inference and First-Order Inference. Explain the concept of 'Lifting' and how it optimizes the inference process.	(16)	BTL4	Analyzing
13	Evaluate the efficiency of Generalized Modus Ponens (GMP) in production systems. How does it simplify the reasoning process compared to standard resolution?	(16)	BTL5	Evaluating
14	Create an ontology for 'Electronic Commerce' that includes categories for products, customers, transactions, and payment methods, along with their hierarchical relationships.	(16)	BTL6	Creating
15	Describe the steps of the Forward Chaining algorithm for definite clauses and illustrate its execution with a suitable example of a rule-based system.	(16)	BTL3	Applying
16	Examine the 'Wumpus World' problem. Represent the rules of the environment using First-Order Logic and explain how an agent can infer the location of a pit.	(16)	BTL4	Analyzing
17	Justify the use of Skolem functions over Skolem constants when eliminating existential quantifiers that are within the scope of universal quantifiers.	(16)	BTL5	Evaluating
18	Formulate a set of First-Order Logic axioms to describe the 'Kinship' domain (e.g., parent, sibling, grandparent) and derive the 'uncle' relationship using inference.	(16)	BTL6	Creating
19	Explain the concept of 'Substitution' and 'Most General Unifier' (MGU). Why is finding the MGU critical for the efficiency of logical inference?	(16)	BTL3	Applying
20	Analyze the complexities involved in Ontological Engineering for large-scale systems. Discuss the trade-offs between expressivity and computational tractability.	(16)	BTL4	Analyzing

UNIT IV – PROBABILISTIC REASONING

Acting under Uncertainty, Basic Probability Notation, Bayes' Rule and its Use, Representing Knowledge in an Uncertain Domain, The Semantics of Bayesian Networks, Exact Inference in Bayesian Networks, Approximate Inference in Bayesian Networks, Temporal Models, Hidden Markov Models

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define uncertainty in the context of artificial intelligence and list two primary sources of it.	BTL1	Remembering
2	Distinguish between prior probability and posterior probability with a simple example.	BTL2	Understanding
3	State the mathematical formula for Bayes' Rule and identify each of its components.	BTL1	Remembering
4	What is meant by a 'Full Joint Probability Distribution' and why is it often impractical for large domains?	BTL2	Understanding
5	Define conditional independence and explain how it simplifies the representation of knowledge.	BTL1	Remembering
6	Explain the topological requirements for a graph to be considered a Bayesian Network.	BTL2	Understanding
7	What is a Conditional Probability Table (CPT) in the context of Bayesian Networks?	BTL1	Remembering
8	Define the 'Markov Blanket' of a node in a Bayesian Network.	BTL1	Remembering
9	Explain the basic idea behind the Variable Elimination algorithm for exact inference.	BTL2	Understanding
10	What is the primary difference between exact inference and approximate inference in probabilistic reasoning?	BTL2	Understanding
11	Define 'Rejection Sampling' and state one of its major limitations.	BTL1	Remembering
12	How does Likelihood Weighting improve upon the standard Rejection Sampling method?	BTL2	Understanding
13	Define a 'Stationary Process' in the context of temporal models.	BTL1	Remembering
14	Explain the 'First-Order Markov Assumption' used in temporal probabilistic models.	BTL2	Understanding
15	What are the two main components of a Hidden Markov Model (HMM)?	BTL1	Remembering
16	Differentiate between the 'Filtering' and 'Prediction' tasks in temporal reasoning.	BTL2	Understanding
17	What is the purpose of the Viterbi algorithm in Hidden Markov Models?	BTL1	Remembering
18	Define 'd-separation' and its role in determining independence in a Bayesian Network.	BTL2	Understanding
19	What is a 'Belief State' in the context of acting under uncertainty?	BTL1	Remembering
20	Briefly explain the concept of Gibbs Sampling as an MCMC method.	BTL2	Understanding

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Apply Bayes' Rule to a medical diagnosis problem: Given the probability of a disease is 0.01, the test sensitivity is 0.95, and the false positive rate is 0.05, calculate the probability that a patient has the disease given a positive test result.	(16)	BTL3	Applying
2	Design a Bayesian Network for a 'Smart Home Alarm System' involving variables for Burglary, Earthquake, Alarm, JohnCalls, and MaryCalls. Provide the network structure and explain the conditional independence assumptions made.	(16)	BTL6	Creating
3	Analyze the semantics of Bayesian Networks by explaining how a joint probability distribution can be factored into a product of local conditional distributions.	(16)	BTL4	Analyzing
4	Evaluate the computational complexity of the Enumeration algorithm for exact inference and explain why it is inefficient for large networks compared to Variable Elimination.	(16)	BTL5	Evaluating
5	Demonstrate the step-by-step process of the Variable Elimination algorithm to compute a query $P(A e)$ in a given 5-node directed acyclic graph.	(16)	BTL3	Applying

6	Compare and contrast Rejection Sampling, Likelihood Weighting, and Gibbs Sampling for approximate inference, highlighting the scenarios where each is most effective.	(16)	BTL4	Analyzing
7	Construct a Hidden Markov Model for a part-of-speech tagging task. Define the states, observations, transition probabilities, and emission probabilities involved.	(16)	BTL6	Creating
8	Derive the update equations for the Forward-Backward algorithm used in smoothing for temporal models.	(16)	BTL4	Analyzing
9	Justify the use of the Markov assumption in modeling dynamic systems and discuss the trade-offs between model accuracy and computational tractability.	(16)	BTL5	Evaluating
10	Develop a complete probabilistic model for a robot navigating in a 1D grid, incorporating transition and sensor models to handle movement and observation uncertainty.	(16)	BTL6	Creating
11	Analyze the impact of 'explaining away' (intercausal reasoning) in a V-structure of a Bayesian Network with a concrete example.	(16)	BTL4	Analyzing
12	Evaluate the effectiveness of the Kalman Filter as a specific type of temporal model for continuous variables and compare it with HMMs.	(16)	BTL5	Evaluating
13	Apply the Viterbi algorithm to find the most likely sequence of hidden states for a given sequence of observations in a weather-prediction HMM.	(16)	BTL3	Applying
14	Explain the concept of 'Acting under Uncertainty' using the Maximum Expected Utility (MEU) principle and provide a decision-making scenario.	(16)	BTL3	Applying
15	Critically examine the challenges of representing knowledge in uncertain domains when the number of variables is large and the dependencies are complex.	(16)	BTL4	Analyzing
16	Formulate a Bayesian Network to represent the causal relationships in a vehicle's cooling system and explain how to perform diagnostic inference on it.	(16)	BTL6	Creating
17	Compare the 'Filtering' process in HMMs with the 'Smoothing' process, explaining why smoothing generally provides more accurate state estimates.	(16)	BTL4	Analyzing
18	Assess the convergence properties of Markov Chain Monte Carlo (MCMC) methods and explain the 'burn-in' period requirement.	(16)	BTL5	Evaluating
19	Apply the concept of 'Noisy-OR' gates to simplify the representation of conditional probability tables in a Bayesian Network with multiple causes for a single effect.	(16)	BTL3	Applying
20	Design a temporal model for a stock market trend analysis, identifying the hidden states, observable indicators, and the transition/sensor models.	(16)	BTL6	Creating

UNIT V – PLANNING AND LEARNING

Classical Planning: Definition, Algorithms for Planning as State-Space Search, Planning Graphs, Hierarchical Planning, Multi-agent Planning, Forms of Learning, Supervised Learning, Learning Decision Trees, Evaluating and Choosing the Best Hypothesis, Introduction to Reinforcement Learning

PART-A (2 Marks)

Q.No.	Questions	BT Level	Competence
1	Define Classical Planning in the context of Artificial Intelligence.	BTL1	Remembering
2	List the four components of a planning problem in PDDL.	BTL1	Remembering
3	Distinguish between progression and regression planning strategies.	BTL2	Understanding
4	What is a 'Mutex' relation in a Planning Graph?	BTL1	Remembering
5	Explain the concept of 'Level Off' in a Planning Graph.	BTL2	Understanding
6	Define Hierarchical Task Network (HTN) planning.	BTL1	Remembering
7	What is the primary difference between a primitive action and a non-primitive action in HTN?	BTL2	Understanding
8	Identify two main challenges unique to Multi-agent Planning compared to single-agent planning.	BTL2	Understanding
9	Define Supervised Learning with a suitable example.	BTL1	Remembering
10	What is 'Inductive Learning'?	BTL1	Remembering
11	State the formula for Entropy as used in Decision Tree induction.	BTL1	Remembering
12	Explain the significance of Information Gain in choosing a split attribute.	BTL2	Understanding
13	What is Ockham's Razor principle in the context of hypothesis selection?	BTL2	Understanding
14	Define 'Overfitting' in machine learning.	BTL1	Remembering
15	What is the purpose of a 'Validation Set' during the learning process?	BTL2	Understanding
16	Define Reinforcement Learning (RL).	BTL1	Remembering
17	What are the three main components of an RL agent's environment interaction?	BTL1	Remembering
18	Differentiate between a 'Policy' and a 'Reward Function' in RL.	BTL2	Understanding
19	What is K-fold cross-validation?	BTL1	Remembering
20	Explain the concept of 'Hypothesis Space' in learning.	BTL2	Understanding

PART-B (16 Marks)

Q.No.	Questions	Marks	BT Level	Competence
1	Examine the STRIPS language for classical planning. Illustrate how a 'Blocks World' problem is represented and solved using state-space search.	(16)	BTL4	Analyzing
2	Compare Forward State-Space Search (Progression) and Backward State-Space Search (Regression) with respect to branching factor and efficiency.	(16)	BTL4	Analyzing
3	Construct a Planning Graph for the 'Spare Tire' problem. Detail the expansion of literal and action levels and identify all possible mutex relations.	(16)	BTL6	Creating
4	Explain the Graphplan algorithm in detail. Discuss how it uses the planning graph to extract a valid plan and handle backtracking.	(16)	BTL3	Applying
5	Elaborate on Hierarchical Planning. Explain the process of 'Decomposition' and how HTN planners manage constraints between sub-tasks.	(16)	BTL3	Applying
6	Analyze the coordination mechanisms in Multi-agent Planning. Discuss how agents handle joint goals and resource contention in a shared environment.	(16)	BTL4	Analyzing

7	Classify and explain the different forms of learning: Supervised, Unsupervised, Semi-supervised, and Reinforcement Learning with real-world applications.	(16)	BTL4	Analyzing
8	Design a Decision Tree using the ID3 algorithm for a given dataset (e.g., 'Play Tennis'). Show step-by-step calculations for Entropy and Information Gain.	(16)	BTL6	Creating
9	Evaluate the methods used for choosing the best hypothesis. Discuss the trade-offs between model complexity and generalization error.	(16)	BTL5	Evaluating
10	Explain the mathematical framework of Reinforcement Learning. Describe the Markov Decision Process (MDP) and the role of the Bellman equation.	(16)	BTL3	Applying
11	Critically analyze the problem of 'Exploration vs. Exploitation' in Reinforcement Learning. Suggest strategies to balance these two aspects.	(16)	BTL5	Evaluating
12	Develop a PDDL domain and problem file for a 'Logistics' scenario involving trucks, airplanes, and packages across different cities.	(16)	BTL6	Creating
13	Discuss the various techniques for evaluating the performance of a learning algorithm, including Learning Curves and Confusion Matrices.	(16)	BTL5	Evaluating
14	Analyze the impact of 'Pruning' (Pre-pruning and Post-pruning) on the performance and size of Decision Trees.	(16)	BTL4	Analyzing
15	Explain how Planning Graphs can be utilized to generate admissible heuristics for A* search in planning problems.	(16)	BTL3	Applying
16	Compare and contrast the 'Total-order' and 'Partial-order' planning approaches. Provide examples where one is preferred over the other.	(16)	BTL4	Analyzing
17	Justify the use of Ensemble methods like Bagging and Boosting in supervised learning to improve hypothesis accuracy.	(16)	BTL5	Evaluating
18	Formulate a solution for a multi-agent navigation task using decentralized planning techniques. Discuss the role of communication.	(16)	BTL6	Creating
19	Explain the concept of 'Inductive Bias'. How does the choice of bias affect the learning capability of a Decision Tree algorithm?	(16)	BTL3	Applying
20	Assess the challenges of applying Classical Planning to real-world domains with uncertainty and continuous time.	(16)	BTL5	Evaluating

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